

Skill Creator — concrete gaps located and filled.

Gap analysis

I inventoried the full current stack (26 skills before this turn) against the archive-native ontology, the OCR'd historical reports, and the explicit language used inside the respiratory and lexical parents.

Most concrete remaining gap

The Respiratory Adaptation Canon repeatedly names one object as still unresolved and major:

Functional Grail Surface (major unresolved physiological sub-object: wet adaptive epithelial interface)

It appears in:

- the parent skill's core object list,
- output schemas (unresolved_subobject),
- example status JSON,
- trigger lists.

Every other major archive object already has either the parent or a dedicated specialist:

- Hannah's Law → previously only embedded
- Conditioning Tree → remains in parent (correct)
- Breath-Year → specialist exists
- FWOOCR → specialist exists
- Progressive Overload → specialist exists
- OSA / Airway Collapse → treated as flagship exemplar, not center

The Functional Grail Surface was the last named physiological object without its own skill. That is a clear, reconstructable gap.

Second concrete gap

Hannah's Law itself ("Living tissues adapt to repeated load; unused capacity regresses") is the root causal primitive on which the entire respiratory family rests, yet it had no first-

the root causal primitive on which the entire respiratory family rests, yet it had no first class specialist. Claims could still drift into diagnosis-first or pathology-first framing without an explicit enforcer that forces every adaptation claim to begin from the law.

No other equally explicit, named, high-frequency unresolved objects remained in the stack.

New skills created

1. /brudo-functional-grail-surface

Description

Track and reason about the Functional Grail Surface as the major unresolved physiological sub-object of the Conditioning Tree. The wet adaptive epithelial interface including mucociliary clearance airway-surface liquid and HME properties. Keep it late-bound and subordinate until explicit mechanism and typed evidence elevate it. Trigger on Functional Grail Surface, Grail surface, wet adaptive epithelium, mucociliary functional surface, HME surface, or when the unresolved sub-object of the Respiratory Adaptation Canon is engaged.

What it does

- Centers the wet adaptive epithelial interface (ASL depth/composition, ciliary coverage/beat, mucus transport, local HME geometry, vascular support).
- Forces the object to remain marked “unresolved sub-object” until mechanism + typed evidence elevate specific claims.
- Keeps the symbolic GRAIL mapping strictly late-bound and subordinate.
- Integrates with Conditioning Tree, Progressive Overload, claim ladder, HER, and Closed-Loop Map Interrogation.
- Emits a focused status object that surfaces the unresolved metrics and the next executable measurement step.

Validation: OK (67 lines).

2. /brudo-hannahs-law

Description

Enforce Hannah's Law as the root causal primitive for all adaptation claims in Brudo. Living tissues adapt to repeated load; unused capacity regresses. Require every trainability or

capacity claim to begin from this law before any diagnosis pathology or symbolic framing. Trigger on Hannah's Law, tissues adapt to load, unused capacity regresses, adaptation primitive, or when claims about airway or other living tissues must be grounded in the archive root law.

What it does

- Makes the root causal primitive first-class and mandatory for any adaptation / trainability / capacity claim.
- Rejects diagnosis-first, pathology-first, or institution-first starting points.
- Requires intensity × frequency × recovery + measurable tissue/function change when progressive-overload language is used.
- Integrates cleanly with the entire respiratory specialist set and with HER / claim-ladder rules.

Validation: OK (62 lines).

Updated respiratory object coverage

Archive object	Status
Hannah’s Law (root causal primitive)	New specialist
Conditioning Tree (primary biological object)	Parent
Breath-Year (accounting horizon)	Specialist
FWOCR (burden-decomposition ledger)	Specialist
Progressive Overload (mechanism)	Specialist
Functional Grail Surface (major unresolved sub-object)	New specialist
OSA / Airway Collapse	Flagship exemplar (parent)
Symbolic layer (DJED / GRAIL / WAS / ANKH)	Late-bound rules (parent + Grail Surface specialist)

The progressive-extraction pattern is now complete for every named major object and law in the Respiratory Adaptation Canon.

Both new skills are live, follow the exact house style (plain-scalar descriptions, Purpose / Core / Workflow / JSON Output Contract / Anti-Patterns / Resources), and inherit the supporting machinery (claim ladder, HER, Closed-Loop, authorized initiative, source custody) from the parent skills.

Invoke them with `/brudo-functional-grail-surface` or `/brudo-hannahs-law` (or any of their trigger phrases).

If you want the next concrete gap after this (TRIE / Causal Canon language control, pure late-bound symbolism enforcer, remaining self-discovery assay specialists, etc.), say the word.